

1. Administrative & Study Identification

Study Title: Substudy 02D: Safety and Efficacy of Pembrolizumab in Combination With Investigational Agents or Pembrolizumab Alone in Participants With Melanoma Brain Metastasis (MK-3475-02D/KEYMAKER-U02) **Official Title:** A Phase 1/2 Open-Label Rolling-Arm Umbrella Platform Design of Investigational Agents With or Without Pembrolizumab or Pembrolizumab Alone in Participants With Melanoma (KEYMAKER-U02): Substudy 02D **Short Title/Acronym:** KEYMAKER-U02 Substudy 02D **Protocol Number:** MK-3475-02D [2023-506315-17-00] **NCT Number:** NCT04700072 **Trial Status:** Completed **Sponsor/Company Name:** Merck Sharp & Dohme LLC (Merck) **Investigational Product:** Pembrolizumab, Lenvatinib, Quavonlimab (MK-1308A) **Phase of Development:** Phase 1, Phase 2 (Integrated) **Disease Area:** Melanoma **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Director

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of investigational agents (percentage of participants experiencing an Adverse Event [AE] and percentage discontinuing treatment due to AEs), and to assess the Objective Response Rate (ORR) in participants with melanoma brain metastasis (MBM). **Secondary Objectives:** To evaluate Duration of Response (DOR), Brain Metastasis Response Rate (BMRR), Brain Metastasis Duration of Response, and Progression-Free Survival (PFS).

2.2 Background & Rationale Melanoma is notorious for spreading to the brain, and patients who develop melanoma brain metastases (MBM) have historically faced poor prognoses. Substudy 02D is part of a larger umbrella platform study testing experimental treatments for melanoma. The goal is to evaluate the safety and efficacy of investigational treatment arms in programmed cell-death 1 (PD-1) naïve or PD-1 exposed participants with MBM, and to identify agent combinations (e.g., PD-1 inhibition via pembrolizumab and VEGF inhibition via lenvatinib) that are superior to the current available treatment options.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Umbrella Platform Design) **Control Type:** Active Comparator / Rolling-arm design **Blinding:** Open-label **Randomization:** Randomized

(Participants are randomly assigned to currently open treatment arms)

3.2 Planned Enrollment & Duration Targeted Enrollment: 56 participants **Number of Sites:** Multicenter global study **Estimated Study Start:** April 2020 **Estimated Primary Completion:** Treatment duration lasts up to approximately 2 years, with an ongoing survival follow-up up to ~30 months.

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Confirmed diagnosis of advanced Stage IV, M1D melanoma with brain metastases. 3. Neurologically asymptomatic from brain metastases. 4. Participants may be PD-1 naïve or PD-1 exposed.

4.2 Exclusion Criteria 1. Presence of symptomatic brain metastases requiring immediate local intervention. 2. Concomitant use of systemic corticosteroids for the management of brain metastases at the time of study initiation. 3. Prior local therapy or whole-brain radiation therapy (WBRT) where applicable/recent.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants receive study medication combinations such as Pembrolizumab + Lenvatinib, or Co-formulated Pembrolizumab/Quavonlimab (MK-1308A) + Lenvatinib. Dosing cycles occur approximately every 3 weeks.

Route of Administration: Intravenous (for immunotherapy antibodies like pembrolizumab/quavonlimab) and Oral (for lenvatinib). **Duration of Treatment:** Up to approximately 2 years, or until disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for advanced melanoma. **Prohibited:** Routine use of systemic corticosteroids for MBM management during the baseline/induction phase, as it confounds the immunotherapy efficacy assessment.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline MRI/Imaging, Eligibility Assessment
Baseline	Day 0	Randomization, First Dose
Interim	Every 3 weeks (first 12 weeks), then every 6 weeks	Safety Labs, Efficacy Assessment (Tumor Imaging), AE monitoring
End of		

Study | Follow-up Phase | Final Vital Signs, Exit Interview, IP Accountability, survival follow-up every 12 weeks via phone |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: 1. Percentage of participants with AEs. 2. Percentage of participants who discontinue study treatment due to AEs. 3. Objective Response Rate (ORR).

Measurement Method: Clinical toxicity grading for AEs. ORR is measured using RECIST 1.1 and Response Assessment in Neuro-Oncology Brain Metastases (RANO-BM) criteria, as assessed by a Blinded Independent Central Review (BICR).

7.2 Secondary & Exploratory Endpoints * Duration of Response (DOR) * Brain Metastasis Response Rate (BMRR) * Brain Metastasis Duration of Response * Progression-Free Survival (PFS)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring through physical exams, blood tests, and clinical evaluation at every scheduled visit. **Serious Adverse Events (SAEs):** Standard reporting timelines (typically 24 hours) as governed by the protocol and regulatory authorities. **Data Monitoring Committee (DMC):** Internal and external safety review to evaluate investigational agent tolerability and determine Recommended Phase 2 Doses (RP2D).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy parameters; Safety Set for all participants receiving at least one dose of the study drug. **Sample Size Justification:** Adaptive umbrella platform design; specific treatment arms are expanded based on the totality of evidence and preliminary efficacy.

Handling of Missing Data: Standard statistical imputation methods as defined by the sponsor's SAP.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular clinical site monitoring, centralized medical review, and BICR evaluation for all tumor imaging data.

Audit Trail: Standardized tracking of electronic Case Report Forms (eCRF) and source document verification.

11. Study Identifiers & Governance

Primary ID: MK-3475-02D **Registry Number:** NCT04700072 **Sponsor/Company:** Merck **Study Title:** Substudy 02D: Safety and Efficacy of Pembrolizumab in Combination With Investigational Agents or Pembrolizumab Alone in Participants With Melanoma Brain Metastasis (MK-3475-02D/KEYMAKER-U02) **Official Regulatory Title:** A Phase 1/2 Open-Label Rolling-Arm Umbrella Platform Design of Investigational Agents With or Without Pembrolizumab or Pembrolizumab Alone in Participants With Melanoma (KEYMAKER-U02): Substudy 02D

12. Clinical Framework (Melanoma Brain Metastasis Specific)

Primary Condition: Advanced Melanoma with Brain Metastases (Stage IV, M1D) **Diagnosis Threshold:** Clinically confirmed MBM, neurologically asymptomatic **Medical Methodology:** Combined immune checkpoint inhibition (anti-PD-1 / anti-CTLA-4) and targeted kinase inhibition (anti-VEGFR/FGFR) **Optimization Target:** Systemic and intracranial tumor response without excessive toxicity

13. Study Procedures & Methodology

Management Pathway: Open-label, multi-arm allocation to evaluate novel combinations. **Education Protocol (HCP):** Site initiation visits and Investigator Brochure (IB) training on expected immune-related adverse events (irAEs). **Education Protocol (Patient):** Informed consent process detailing potential immunotherapy/targeted therapy toxicities. **Quality Audit Mechanism:** Centralized independent radiological review (BICR) using RANO-BM.

14. Observation & Follow-Up Schedule

Total Duration: Up to ~30 months **Onsite Visit Cadence:** Every 3 weeks for the first 12 weeks, followed by every 6 weeks. **Remote Monitoring Frequency:** During the follow-up phase, participants are contacted by phone approximately every 12 weeks to inquire about their health status. **Checklist Review Interval:** Safety and efficacy review at each dosing cycle.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				